

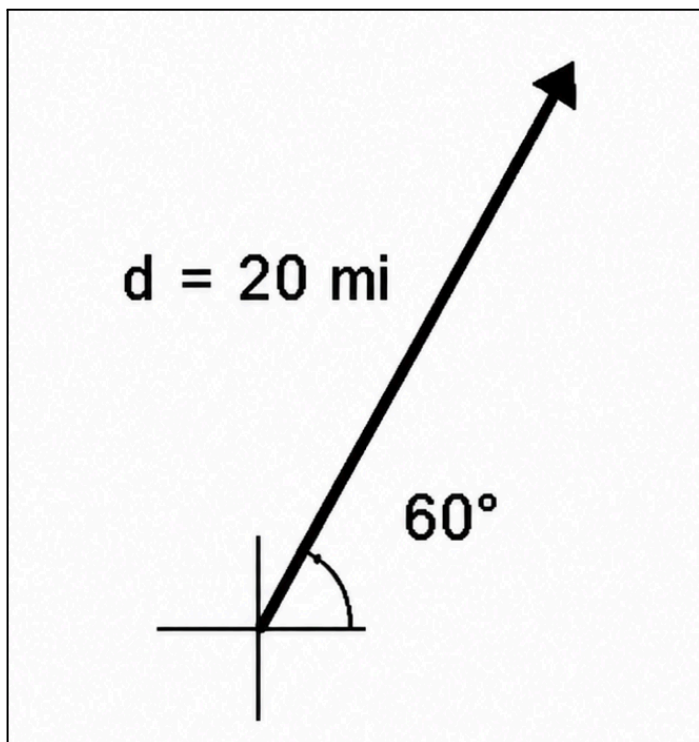
1.) Find the equilibrant of the 2 vectors given below.

$$P = \langle -2, 10, -5 \rangle$$

$$Q = \langle 3, 2, 1 \rangle$$

- A.  $\langle 1, 12, -4 \rangle$
- B.  $\langle -1, -12, 4 \rangle$
- C.  $\langle -6, -20, 5 \rangle$
- D.  $\langle 6, 20, -5 \rangle$

Answer: B.  $\langle -1, -12, 4 \rangle$



2.) Which of the following best describe the direction of the vector illustrated in the figure above?

- A. 60 degrees West of North
- B. 30 degrees East of North
- C. 60 degrees East of North
- D. 30 degrees North of East

Answer: B. 30 degrees East of North

3.) A projectile fired at an angle of  $45^\circ$  above the horizontal. Assume that air resistance is not significant. While the projectile is in flight, its horizontal speed remains constant.

- A. True
- B. False

Answer: A. True

4.) Which of the following vectors is a unit vector?

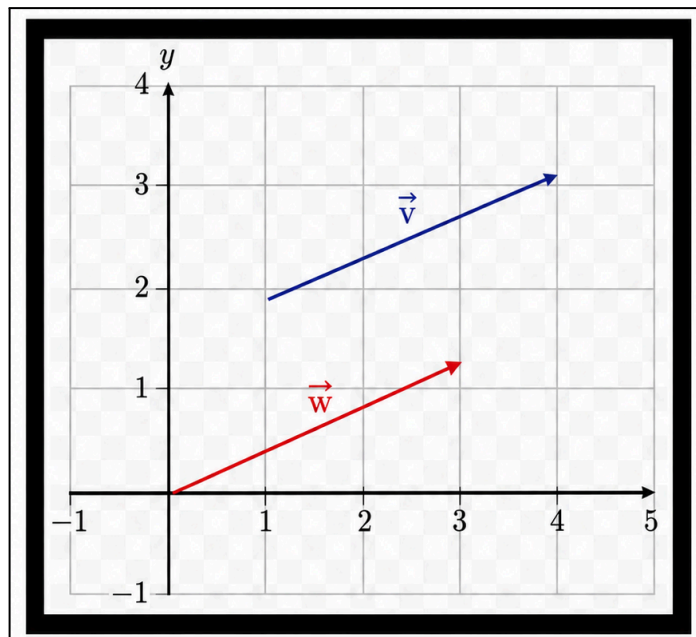
- A.  $\langle -1, -1 \rangle$
- B.  $\langle 1, 0, 1 \rangle$
- C.  $\langle -1, 0 \rangle$
- D.  $\langle 1, 1 \rangle$

Answer: C.  $\langle -1, 0 \rangle$

5.) Given the vectors  $M = 2i - 3j$  and  $N = -3i + 2k$ , what is the resultant?

- A.  $i - j$
- B.  $-i - k$
- C.  $i + 3j - 2k$
- D.  $-i - 3j + 2k$

Answer: D.  $-i - 3j + 2k$



6.) Which of the following is NOT TRUE about the two vectors illustrated in the figure above?

- A. The vectors are of equal magnitudes.
- B. The vectors are coinciding each other.
- C. The vectors are parallel to each other.
- D. The vectors are located in the first quadrant.

Answer: B. The vectors are coinciding each other.

7.) What is the y-component of vector C if  $C = -12j + 2i - 5k$ ?

- A. 2
- B. -12
- C. 12
- D. -5

Answer: B. -12

8.) Given the statements below, which is/are ALWAYS true?

- A. The cross product of 2 parallel vectors is always negative.
- B. The dot product of perpendicular vectors is always maximum.
- C. The vector sum of 2 opposite vectors is always zero.

- A. Statement C only
- B. Statements A and B only
- C. None of the statement
- D. Statements A, B and C

Answer: A. Statement C only

**9.) Which of the following will result to a quantity with no direction?**

- A. equilibrant of 2 vectors
- B. cross product of 2 vectors
- C. resultant of 2 vectors
- D. dot product of 2 vectors

**Answer: D. dot product of 2 vectors**

**10.) A car accelerates from 2 m/s to 5 m/s in 10 seconds, and then continues at this constant speed for another 10 seconds. What is the total distance travelled by the car?**

- A. 50m
- B. 60m
- C. 72m
- D. 85m

**Answer: D. 85m**

**11.) The only force that affects the motion of a projectile is**

- A. Air Resistance
- B. Tension
- C. Gravity
- D. Friction

**Answer: C. Gravity**

**12.) A car travelling at 20 m/s rounds a curve so that its centripetal acceleration is 5 m/s<sup>2</sup>. What is the radius of the curve?**

- A. 80 m
- B. 8 m
- C. 640 m
- D. 40 m

**Answer: A. 80 m**

**13.) Vector A and Vector B has magnitudes of 15 and 10, respectively. If the angle between these vectors is 35°, what is the magnitude of the cross product of vectors A and B?**

- A. 0
- B. 123
- C. 86
- D. 105

**Answer: C. 86**

**14.) Which of the following is an indication of a body having acceleration?**

- A. a change in direction
- B. a decrease in speed
- C. an increase in speed
- D. all of the choices

**Answer: D. all of the choices**

**15.) A projectile is fired from a gun and has initial horizontal and vertical components of velocity equal to 30 m/s and 40 m/s, respectively. Determine the magnitude of the initial velocity of the projectile.**

- A. 50 m/s
- B. 80 m/s
- C. 60 m/s
- D. 70 m/s

**Answer: A. 50 m/s**

**16.) Starting from rest, a particle confined to move along a straight line is accelerated at a rate of 2.0 m/s<sup>2</sup>. Which one of the following statements accurately describes the motion of this particle?**

- A. The particle travels 2.0 m during each second.
- B. The particle travels 2.0 m only during the first second.
- C. The speed of the particle increases by 2.0 m/s during each second.
- D. The acceleration of the particle increases by 2.0 m/s<sup>2</sup> during each second.

**Answer: C. The speed of the particle increases by 2.0 m/s during each second.**

**17.) You can't decide where to eat so you flipped a coin with an initial velocity of 10 m/s. How long is the coin in the air before it lands on you palm?**

- A. 9.80 s
- B. 1.02 s
- C. 2.04 s
- D. 0.54 s

**Answer: C. 2.04 s**

**18.) A race car moving with a constant speed of 60 m/s completes one lap around a circular track in 50 s. What is the magnitude of the acceleration of the race car in m/s<sup>2</sup>?**

- A. 5.5 m/s<sup>2</sup>
- B. 4.5 m/s<sup>2</sup>
- C. 7.5 m/s<sup>2</sup>
- D. 6.5 m/s<sup>2</sup>

**Answer: C. 7.5 m/s<sup>2</sup>**

**19.) A person, starting from the origin, walks 8 km east during first day, and 5 km west during the next day. What is the net displacement of the person from the initial point in two days?**

- A. 13km, west
- B. 6km, east
- C. 10km, west
- D. 3km, east

**Answer: D. 3km, east**

20.) Which of the following is used to determine the direction of a cross-product result?

- A. Pythagorean Theorem
- B. Rule of Thumb
- C. Head to Tail Method
- D. Right Hand Rule

**Answer: D. Right Hand Rule**

21.) The vector  $A = \langle 1, 0, -1 \rangle$  is a vector in the \_\_\_\_\_ plane.

- A. xy plane
- B. xz plane
- C. yz plane
- D. origin

**Answer: B. xz plane**

22.) Your jumps straight upwards with a velocity of 20 m/s. What is his displacement after 3 seconds?

- A. 24.91 m
- B. 20.40 m
- C. 15.90 m
- D. 4.51 m

**Answer: C. 15.90 m**

23.) The magnitude of the first vector is twice that of the second vector. What is the cross product of these two vectors if they are directed opposite to each other?

- A. 0
- B. 1
- C. Can not be determined with the given
- D. 2

**Answer: A. 0**

24.) An object starts from rest and accelerates uniformly in a straight line in the positive x direction. After 11 seconds, its speed is 70.0 m/s. Determine the acceleration of the object.

- A.  $6.4 \text{ m/s}^2$
- B.  $-6.4 \text{ m/s}^2$
- C.  $-3.5 \text{ m/s}^2$
- D.  $3.5 \text{ m/s}^2$

**Answer: A.  $6.4 \text{ m/s}^2$**

25.) A golf ball is hit at an angle of  $45^\circ$  above the horizontal. What is the acceleration in  $\text{m/s}^2$  of the golf ball at the highest point in its trajectory? Neglect the effect of friction.

- A. 0
- B. 9.8 to the left
- C. 9.8 downwards
- D. 9.8 upwards

**Answer: C. 9.8 downwards**

26.) Find the magnitude of  $A = 8i - 12j + 15k$ .

- A. 16.1
- B. 20.8
- C. 32.2
- D. 24.7

**Answer: B. 20.8**

27.) A ball is thrown horizontally from the top of a building 0.10 km high. The ball strikes the ground at a point 65 m horizontally from and below the point of release. What is the speed in m/s of the ball just before it strikes the ground?

- A. 57 m/s
- B. 14 m/s
- C. 32 m/s
- D. 46 m/s

**Answer: D. 46 m/s**

28.) From his house, Brimstone walked 200m,  $30^\circ$  N of E. Then he walked an additional 300m towards West. What is his displacement from his house to the destination?

- A.  $-300i + 200j$
- B.  $173i + 127j$
- C.  $-127i + 100j$
- D.  $-173i + 100j$

**Answer: C.  $-127i + 100j$**

29.) A 5-kg watermelon and a 200-gram apple were dropped at the same time from a 500-meter building. Ignoring air resistance, which of the two will reach the ground first?

- A. Both will reach the ground at the same time
- B. Neither will reach the ground
- C. Apple
- D. Watermelon

**Answer: A. Both will reach the ground at the same time**

30.) Find the magnitude and direction / angle of inclination of the vector in 2D space given below.

$$B = 3i + 3j$$

- A. 9, 38 degrees
- B. 4.24, 38 degrees
- C. 4.24, 45 degrees
- D. 9, 45 degrees

**Answer: C. 4.24, 45 degrees**

31.) Gravity makes things fall on earth and has a constant rate of  $9.8 \text{ m/s}^2$ .

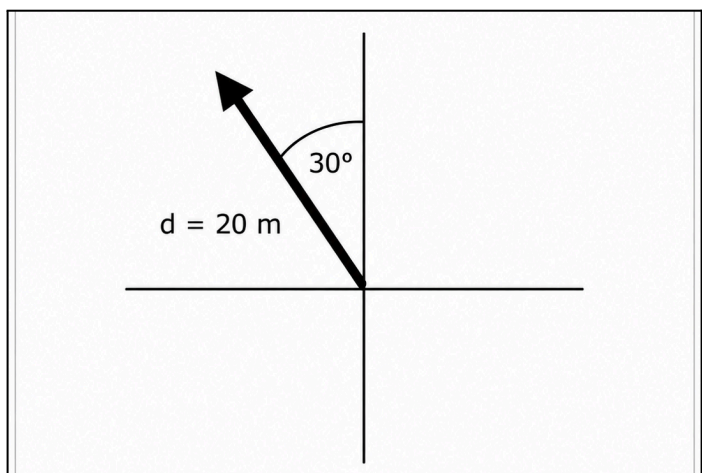
- A. False
- B. True

**Answer: B. True**

32.) A projectile is directed toward the sky with an angle from the level ground. It reaches a range of 10 m in 10s. How long will it reach its maximum height above the ground?

- A. 10s
- B. 5s
- C. 20s
- D. 15s

Answer: B. 5s



33.) Which of the following best describe the direction of the vector illustrated in the figure above?

- A. 30 degrees West of North
- B. 30 degrees North of East
- C. 30 degrees North of West
- D. 60 degrees West of North

Answer: A. 30 degrees West of North

34.) Which of the following is the unit vector representation of the vector <-3, -2, -5>?

- A. <0.49, 0.32, 0.81>
- B. <0.49, -0.32, 0.81>
- C. <-0.49, -0.32, -0.81>
- D. <-0.49, 0.32, 0.81>

Answer: C. <-0.49, -0.32, -0.81>

35.) The unit of centripetal acceleration is \_\_\_\_\_.

- A. m/s
- B. rpm
- C. Hertz
- D. m/s<sup>2</sup>

Answer: D. m/s<sup>2</sup>

36.) Which of the following is a vector quantity?

- A. Distance
- B. Mass
- C. Acceleration
- D. Speed

Answer: C. Acceleration

37.) When an object is released from rest and falls in the absence of air resistance, which of the following is true concerning its motion?

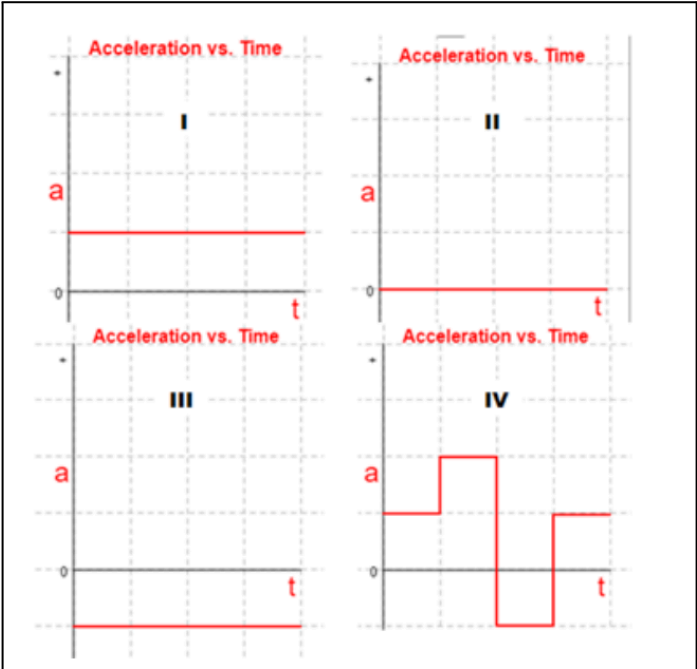
- A. Neither its acceleration nor its velocity is constant.
- B. Its acceleration is constant.
- C. Both its acceleration and its velocity are constant.
- D. Its velocity is constant.

Answer: B. Its acceleration is constant.

38.) What is the dot product of two vectors that are equal in magnitude and with 45° angle in between?

- A. 0.71
- B. 2
- C. Cannot be determined with the given
- D. 1.41

Answer: C. Cannot be determined with the given



39.) Which of the following motion graphs best represents a free-falling body?

- A. II
- B. I
- C. IV
- D. III

Answer: D. III

40.) Given the vectors below. What is the angle between vectors C and D?

C = -3i - 3j  
D = -5i - 5j

- A. 0
- B. 45
- C. 180
- D. 90

Answer: A. 0